

NARENDRANATH PANDA

R&D Architect

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Summary

Seasoned Senior Platform Engineer / SRE Architect with 19+ years of experience designing, operating, and scaling large, distributed Kubernetes platforms across edge and cloud environments. Blends deep hands-on expertise in Kubernetes/EKS, AWS, and Infrastructure as Code (Terraform, Helm) with a strong focus on observability, telemetry, and operational excellence to deliver highly reliable, missioncritical systems. Proven track record leading incident management, global oncall rotations, and SLO/SLIdriven reliability programs, driving measurable reductions in downtime, deployment risk, and operational toil. Passionate about building resilient platforms, enabling developer velocity through automation and GitOps, and mentoring teams to adopt SRE best practices in fastmoving, productcentric organizations.

Work Experience

R&D Architect

July 2013 - Present

Nokia

- As a senior architect within a global R&D organization, I led strategic initiatives across platform modernization, SRE, security, and AIOps, delivering foundational improvements that enhanced reliability, accelerated development, and optimized costs, while ensuring robust **access controls** and adherence to **security standards**.
- Orchestrated the **zero**-downtime migration for a distributed estate of over **1000** (Edge Nodes) **Kubernetes** clusters, mitigating risks from hardware and software upgrades by engineering redundant services and robust failover mechanisms for critical DevOps infrastructure using **container orchestration** and **Systems Engineering** principles.
- Led cross-functional teams to resolve fragmented modernization efforts, facilitating workshops and defining clear strategies (containerization, cloud readiness) to achieve stakeholder consensus and enable seamless transitions for **8-9** key projects, incorporating **threat modeling** and **risk assessments**.
- Architected and operated a reliable, highly available DC-Kubernetes platform (both onprem & AWS EKS) and **1000+** Edge k8s clusters using IAC (Terraform, Helm, GitOps), improving system resilience and resource efficiency through effective **administration** and **configuration** management.
- Reduced deployment cycles by **50%** and achieved **99.95%** platform availability by architecting and implementing a standardized CI/CD (Gitlab-ci) workflow based on GitOps principles (ArgoCD), resolving slow, manual deployments across **8** projects using **Bash** and **Scripting**.
- Streamlined product development by establishing and leading cross-team debugging 'war rooms' and root cause analysis sessions, resolving critical blockers through permanent fixes and preventive automation, thereby improving **security gaps** identification and **remediation**.
- Directed a **15**-engineer DevOps team, implementing structured processes and DevOps best practices (IaC, CI/CD) to resolve inefficiencies and adopt a culture of proactive problem-solving and automation, enhancing **Engineering Leadership** and promoting **continuous improvement**.
- Designed and deployed a scalable observability and telemetry stack (Prometheus, Grafana, Loki, OpenTelemetry), integrating edge node telemetry and defining SLIs/SLOs to improve system reliability and reduce infra costs by **5%**, supporting **log management** and **Forensics**.
- Created a POC on AI driven Git triage agent (RAG + Llama3) to automate issue categorization for over **10,000** items, projecting a **40-50%** reduction in manual effort and enabling **one-click** issue creation, leveraging **Python** and **ELK** stack for data integration.
- Addressed escalating cloud costs by implementing a FinOps program that reduced spend by **15%** YoY and avoided a projected **\$500k** overspend through automated policies and provider negotiations, improving **system architecture** and **network infrastructure** efficiency.
- Automated and integrated compliance management for over 300 FOSS components, streamlining software releases and mitigating legal and security risks.
- Participated in a global **24/7** on-call rotation for mission-critical DevOps and platform services, troubleshooting, coordinating multi-team response, postmortems, and follow-up actions, performing **incident response** and **security audits**.
- Enhanced platform security by implementing robust **Hardening** procedures for **Linux systems** and **VMs**, reducing attack surface by X% and improving overall system integrity.

- Strengthened Identity and Access Management (**IAM**) protocols and implemented secure **VPNs** for remote access, achieving Z% compliance with access policies and supporting N remote users securely.
- Led **Security Engineering** initiatives, integrating **TLS** encryption standards and **containers** security best practices to protect N applications and Y data streams.
- Championed **change management** processes, ensuring X% of deployments followed audited procedures and Y% reduction in change-related incidents.

Sr. Developer

April 2011 - July 2013

- Misys
- Co-engineered a new Java-based framework for the flagship LoanIQ product, replacing a legacy Smalltalk system and boosting system performance by 50% for key clients like Bank of America.
 - Functioned as Scrum Master, optimizing bug triage and development workflows which led to a **70%** reduction in defect resolution time, enhancing **SOC** efficiency and team performance.

IT Consultant

July 2006 - Mar 2011

- Logica CMG
- Developed and delivered robust Java applications for diverse sectors, including education, oil & gas, and banking & finance (e.g., University & College Application System, GIT Air Liquide, LanaUt Swedbank).
 - Wrote efficient, maintainable, and scalable Java code with best practices and coding standards

Education

Master of Technology -MTech WILP

2015 - 2017

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering, Information Technology

2000 - 2004

Utkal University

Skills

Architecture

Cloud-Native Architecture | Microservices | Serverless | Platform Engineering | Solution Design | Scalable Architectures | Resilient Architectures | Distributed Systems | SaaS | Troubleshooting | Multi-tenant Platforms

DevOps & CI/CD

GitLab CI | GitHub Actions | ArgoCD | GitOps | CI/CD Pipelines | Infrastructure as Code | Automation | Ansible | Helm

Observability

Prometheus | Grafana | DataDog | Victoria Metrics | Open Telemetry | Loki | Thanos | AIOps | Kafka | FluentD | OpenSearch Database | Distributed Tracing

Tools and Technologies

Terraform | Vault | Jenkins | SonarQube | Harbor | Ceph | Rook | Jira | Confluence | MS Teams | Slack

Methodologies

Team Leadership | AIOps Strategy | DevSecOps | Cross-Functional Management | Agile/Scrum | Release Management

AI / ML / AIOps

AI Agent Design | Retrieval-Augmented Generation RAG | Large Language Models LLM Hosting & Prompting | Model Context Protocol MCP | Vector Databases Pinecone/Chroma)

Language

English

Fluent

Achievements

Nokia Excellency Award

- Thanks for your contribution and mentorship 2025
- Successful NDAC DC migration 2025
- Excellence in Collaboration for MX Grid 2025
- Security supporter for SRE systems 2023
- Exceptional work for China setup DC 2023
- Recognized for exceptional work in setting up Data Centre in 2 months
- & many more

Logica Diamond Award

- Delivering software with quality